DE LA SALLE HOLY CROSS COLLEGE HIGH

Total Marks: 130

Time: 120 minutes

Examiner: Miss B Hannweg Moderator: Mrs D Webb

Name:

Instructions:

1. This question paper consists of 12 pages. Please ensure you have all pages.
2. There are two sections: Section A: 53 marks

   Section B: 77 marks

   Ensure you answer ALL the sections.
3. Section A is to be answered in the answer booklet given and section B on the folio paper provided.
4. Ensure your answer paper has your name on it.
5. It is in your own interest to write neatly and legibly.
6. **You may not use Tippex.**

Section	A					B				
Question number	1	2	3	4	5	6	7	8	9	10
Mark										
Total										

SECTION A

Question 1 **[10]**

1. Four possible answers are given for each of the following questions. Select the most correct answer and write the letter corresponding to the question in the table below.

Question number	1.1.	1.2.	1.3.	1.4.	1.5.
Answer					

1.1. Facilitated diffusion through a cell membrane:
A. requires energy and goes against the concentration gradient.
B. is the movement of small molecules into and out of the cell through the pores between phospholipids.
C. does not require energy and involves the net movement of molecules from a high to a low concentration.
D. does not involve protein carriers. (2)

1.2. Which of the following processes takes place during cellular respiration?
A. Splitting of the water molecules
B. Liberation of oxygen
C. ADP is changed into ATP
D. CO_2 is taken in (2)

1.3. The primary function of the cell nucleus is:
A. Storage of nutrients.
B. Respiration.
C. Cell functioning.
D. Reproduction. (2)

1.4. The small projections found on a plant root that help with the uptake of nutrients is called:
A. Hair-like follicles
B. Root hairs
C. Root cells
D. Projectiles (2)

1.5. The smallest, most specific category of classification is:
A. Family
B. Species
C. Kingdom
D. Order (2)

Question 2 **[10]**

2. Write the correct biological term for the following definitions:

2.1. The phase of mitosis where the chromosomes line up in the middle of the cell. (2)

2.2. The process our body carries out to make energy. (2)

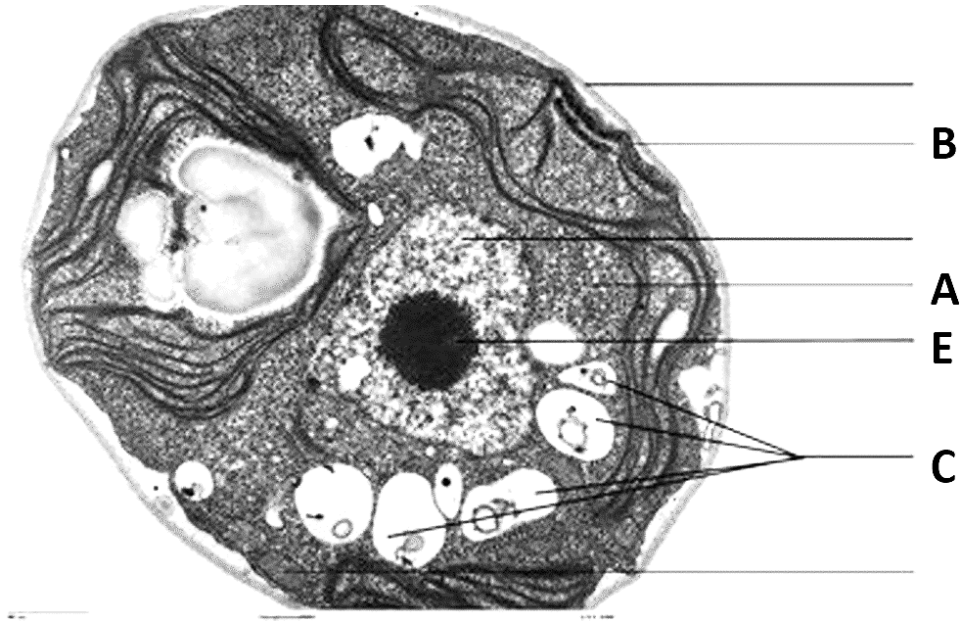
2.3. The age of the dinosaurs. (2)

2.4. Choosing a specific phenotypic trait to duplicate. (2)

2.5. A visual representation of the path of energy exchange through feeding between organisms. (2)

Question 3 **[10]**

3. Study the following diagram and answer the short questions that follow. Include all relevant information.



Adapted from: https://intl.siyavula.com/read/science/grade-9/cells-as-the-basic-units-of-life/01-cells-as-the-basic-units-of-life?id=toc-id-5

3.1. This diagram is a representation of: (2)

3.2. Label E is showing: (2)

3.3. What is the function of label C? (2)

3.4. Label B allows all substances into a cell. True or false? (2)

3.5. Label A represents: (2)

Question 4 **[10]**

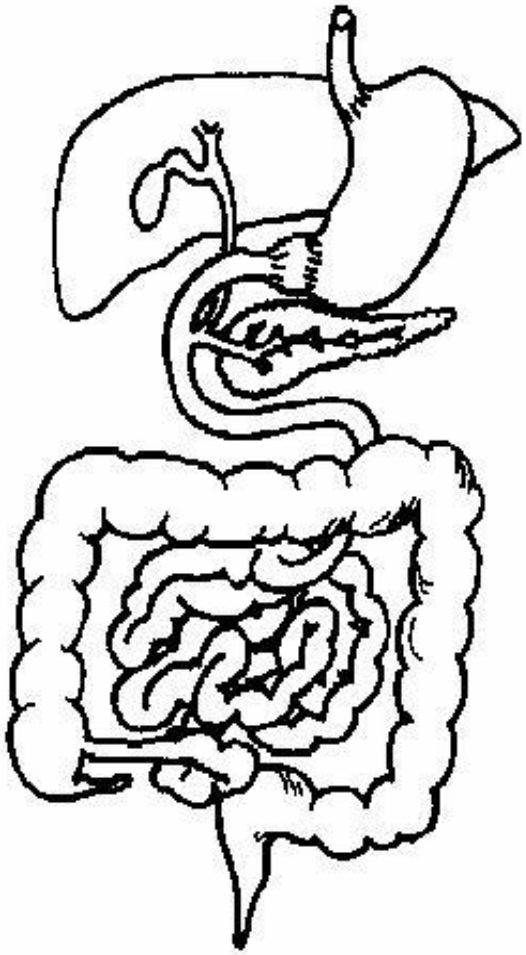
4. Match the defintions in column B with the words in column A. Write only the correct letter next to the word.

Column A	Correct letter	Column B
4.1. Habitat		A. The area or environment where an organism or ecological community normally lives or occurs.
4.2. Biotic		B. Liquid content of the mitochondrion.
4.3. Niche		C. The passive movement of molecules or particles along a concentration gradient, or from regions of higher to regions of lower concentration.
4.4. Stroma		D. The ecological role and space that an organism fills in an ecosystem.
4.5. Stromatolites		E. All plants, animals, fungi, protists and bacteria within an ecosystem.
		F. A major constituent of the fossil record made of Cyanobacteria as well as other prokaryotes.
		G. Liquid content of the chloroplast.

Question 5 **[13]**

5. Add the following labels to the diagram of a mammal's gut below. You may add labels on the left or right of the diagram, for clarity.

pancreas; caecum; stomach; rectum; gall bladder; bile duct; pyloric sphincter; small intestine; colon; liver; oesophagus;



(10)

Adapted from: https://tartrerepub.blogspot.com/2018/08/human-body-diagram-unlabeled.html

5.1. If you have a problem with the roduction of insulin, indicate on the diagram with an **A** where this will occur. (1)

5.2. Acid reflux is a common phenomenon. Indicate with a **B** where the problematic valve is that causes acid reflux. (1)

5.3. Constipation may occur of you follow the incorrect diet. Indicate with a **C** where this will occur. (1)

SECTION B

Instructions:

1. Section B is to be answered on the folio paper provided.
2. Ensure your answer papers have your name on them.
3. It is in your own interest to write neatly and legibly.
4. Leave a ruled margin on both the right and lefthand side of your page.
5. **You may not use Tippex.**

Question 6 [15]

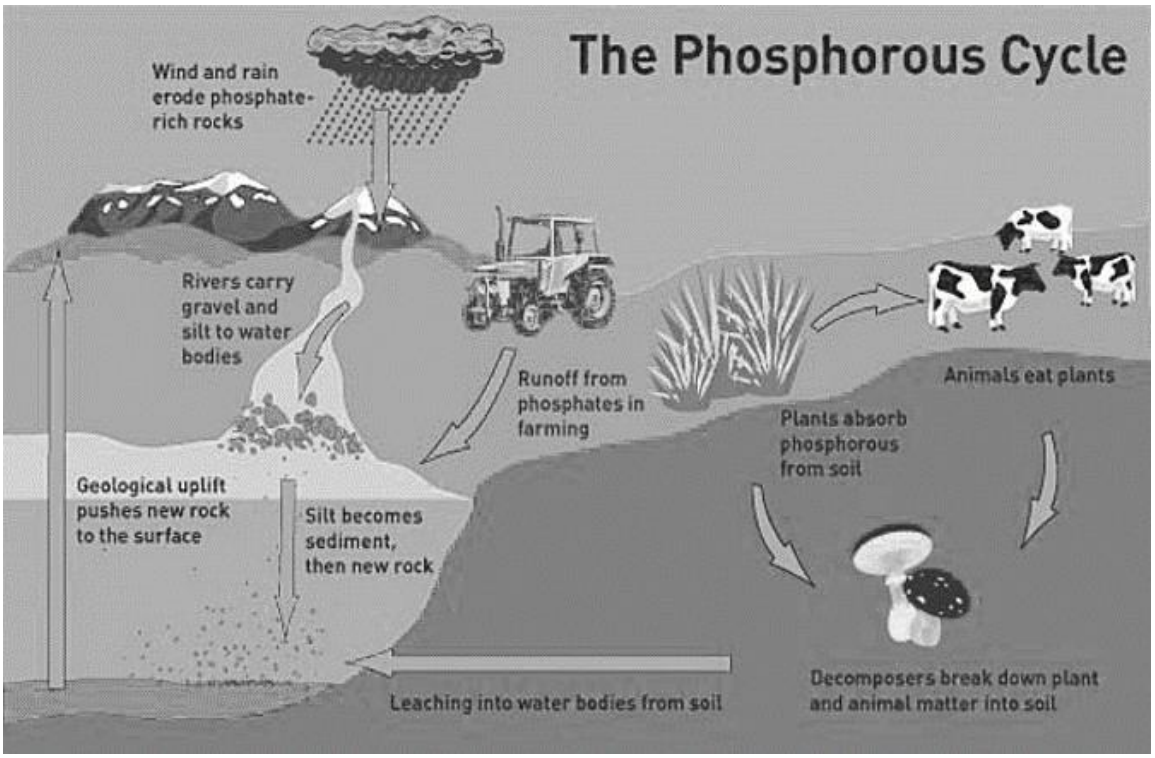
6.1. Phosphorus is needed to make ATP, DNA and phospholipids. Name the organelle or structure within the cell:

6.1.1. where ATP is manufactured. (1)

6.1.2. where DNA is found. (1)

6.1.3. that is made up of phospholipids. (1)

6.2. Below is a diagram of the phosphorus cycle.



Adapted from: https://socratic.org/questions/how-do-humans-interfere-in-the-phosphorus-cycle

6.2.1. Animals obtain their phosphorus in the same way that they obtain carbon.

How do they obtain their phosphorus? (1)

6.2.2. Explain the role of the decomposers (bacteria and fungi) in the phosphorus cycle. (2)

6.2.3. Give two ways in which the role of PLANTS in this cycle differs from their role in the WATER cycle. (2)

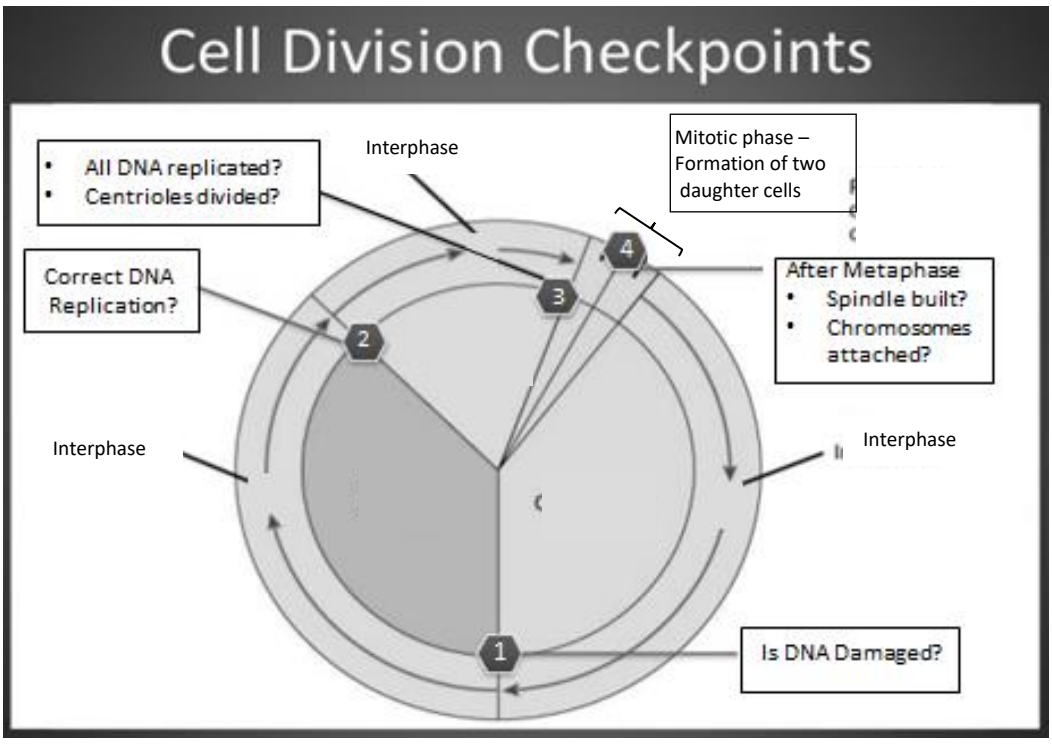
6.2.4. Explain why wind and rain are important in the phosphorous cycle. (1)

6.2.5. What is the importance of nutrient cycles? (3)

6.2.6. Do humans have an impact on all four nutrient cycles? Explain. (3)

Question 7 **[16]**

7. The diagram below shows four different "checkpoints" at which a cell will "check" that mitosis is occurring correctly. These checkpoints help to control mitosis. Each checkpoint is represented by a number (1-4) in the diagram below.



Adapted from: https://quizlet.com/gb/264509129/biology-chapter-6-cell-cycle-checkpoints-diagram/

7.1. Name the longest phase of the cell cycle. (1)

7.2. Why is it essential that all the DNA is replicated prior to mitosis? (2)

7.3. Explain why mitosis is so important to living organisms. (4)

7.4. What happens if there is an error in the duplication of DNA? (2)

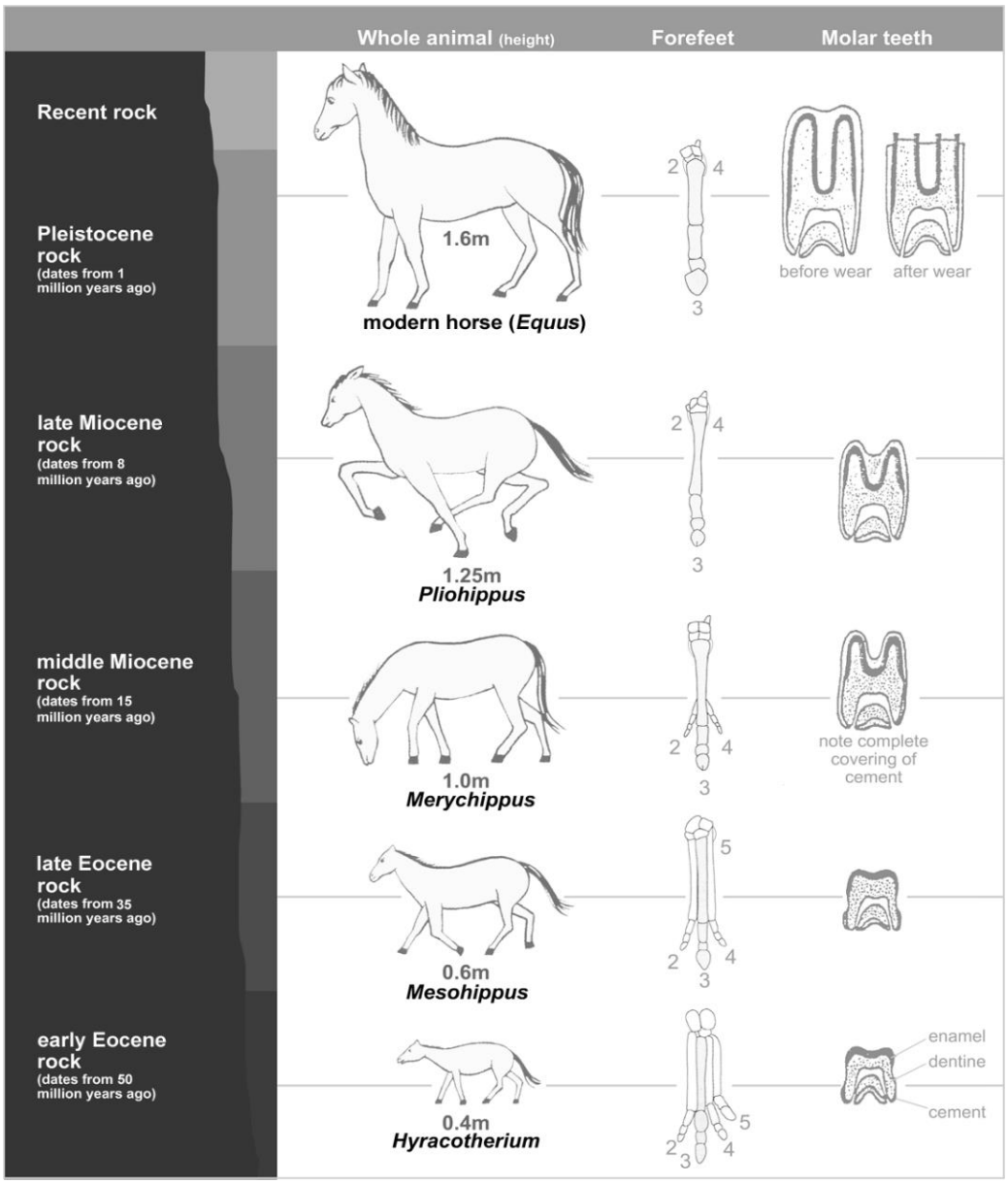
7.5. Draw and label a diagram of a cell containing four chromosomes in the phase

following metaphase. (4)

7.6. Are daughter cells created in mitosis an example of haploid or diploid replicates of the parent cells? Explain. (3)

Question 8 **[11]**

8. Study the representation related to the evolution of horses which shows the fossils of organisms found in rocks from different geological eras.



Adapted from:https://en.wikipedia.org/wiki/Evolution_of_the_horse#/media/File:Horseevolution.png

8.1. Name the species which is the second oldest fossil. (1)

8.2. How long ago did the first horse ancestor live on earth? (1)

8.3. During the evolution of the horse several changes within the forefoot has occurred.

8.3.1. Which number parts have disappeared alltogether? (1)

8.3.2. Which species woud have had three toes touching the ground? (1)

8.4. What is the difference in age between *Merychippus* and *Equus*? (2)

8.5. Why is this type of dating referred to as relative dating? (2)

8.6. Suggest what the chemical conditions of Earth would have been when *Hyracotherium* first evolved and what this would mean for its survival. (3)

Question 9 **[26]**

9. Read the article below and answer the questions which follow:

Mangroves May Save Coral Reefs

Adapted: http://plants-and-animals/mangroves-may-save-coral-reefs. October 2014

Coral reefs represent one of the planet's most endangered ecosystems, with at least three-quarters considered under threat. Overfishing and run-off from land account for much of the problem, but even when local protection is provided, high temperatures and changing ocean chemistry can be enough to destroy reefs on their own.

With a quarter of marine biodiversity on the line, suggestions as radical as covering entire reefs in shade cloth or releasing alkaline substances have been proposed. However, at Hurricane Hole in the Virgin Islands Coral Reef National Monument, the corals have found their own answer.
The roots of red mangrove (*Rhizophora mangle*) plants reach down to the seafloor in the brackish waters of tropical estuaries

The mangroves provide shade during hot spells that cause corals to bleach, a sign of stress that can often precede death. The benefits extend beyond shade, however. The water around the mangrove roots are more alkaline than those in open water, allowing coral larvae to settle and form calcium carbonate skeletons.

Mangroves provide no assistance to offshore reefs, and are under widespread threat themselves. Nevertheless, their conservation (already valued for protection against flooding and carbon storage) could help maintain coral species that might one day be a seed to re-establish destroyed reef populations.

9.1. Name the producer that has the potential to save coral reefs. (1)

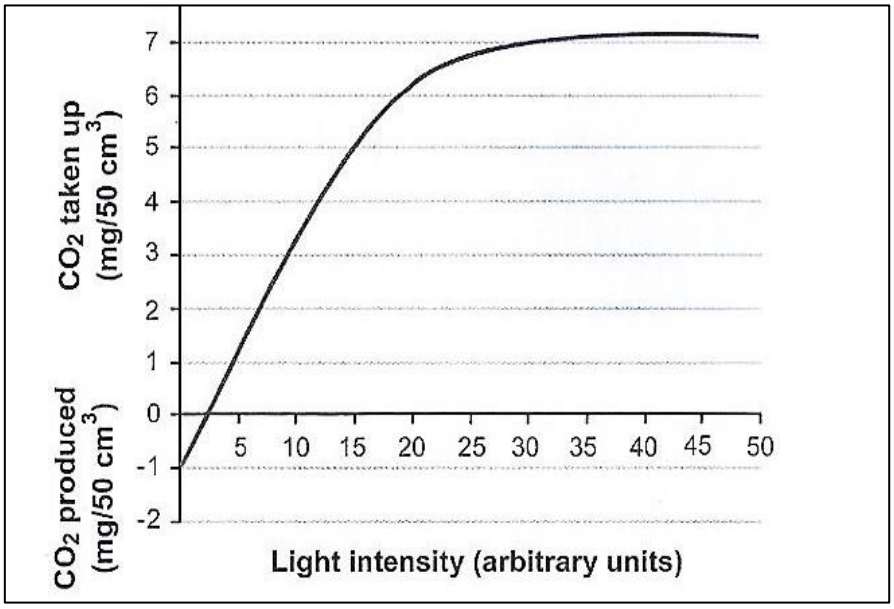
9.2. Identify TWO ways in which the producer mentioned in 9.1 helps to preserve coral reefs. (2)

9.3. Use the information in the passage and your own knowledge to explain THREE ways in which the statement 'Mangroves provide no assistance to offshore reefs' would **NOT** be considered an accurate account of the mangroves. (3)

9.4. Draw a food pyramid of numbers to show FOUR trophic levels of an aquatic ecosystem. (4)

9.5. What is the role of mangrove trees in the carbon cycle? (2)

9.6. The diagram below shows the results of an investigation that was conducted to determine the relationship between light intensity and the release and uptake of Carbon dioxide of a mangrove tree.



9.6.1. Within which range of light intensities is carbon dioxide released? (1)

9.6.2. At a light intensity of above 25 units, the amount of carbon dioxide absorbed remains the same. Provide an explanation for this observation. (2)

9.6.3. Draw a labelled diagram to show the structure of a chloroplast. (include FOUR labelled parts). (6)

9.7. Some species of mangroves have pneumatophores, which are above-ground roots. These are filled with spongy tissue and peppered with small holes that offer structural support and allow oxygen to be transferred to the roots trapped below ground in the anaerobic (low oxygen) soils.

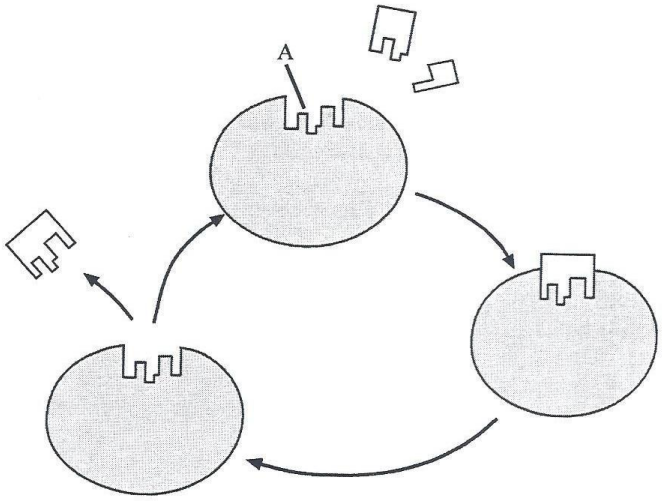
Cellular Respiration is an essential process in mangroves. Name the <u>products</u> of this process under the following conditions:

a. in the presence of oxygen (aerobic respiration) (3)

b. with limited or no oxygen being available (anaerobic respiration) (2)

Question 10 **[9]**

10. The following diagram represents the "lock and key" model which may be used to explain the way in which an enzyme works.



Adapted from: https://www.researchgate.net/figure/Lock-and-key-model-of-enzymatic-action_fig1_335505687

10.1. Name part A. (1)

10.2. Give TWO properties of enzymes that are SHOWN IN THE DIAGRAM. (2)

10.3. Describe how enzymes speed up the rate of a chemical reaction. (4)

10.4. Explain why increasing the temperature above the optimum causes the rate of areaction to decrease. (2)

THE END!